

1 Overview of the course

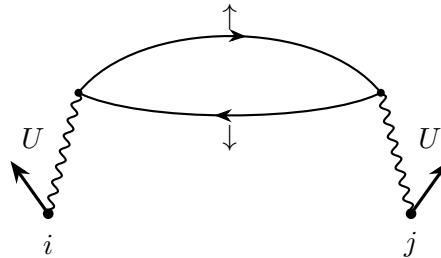
Overall theme of the course.

- (1) In the Ising and Heisenberg models both the spins (localized) and the spin–spin interactions are *given* as input.
- (2) In itinerant-electron models (e.g. the Hubbard model) both the local moments and the interactions between them are *emergent*: they follow from the electronic Hamiltonian.
- (3) Magnetic interactions are mediated by the exchange of the electron–hole (particle–hole) propagator.
- (4) By interpolating properly to the strong-coupling limit, the itinerant-electron approach allows one to study the full range of the interaction strength U/t .

Exchange of the particle–hole propagator. General background for this lecture can be found in [Auerbach \(1994\)](#); [Fazekas \(1999\)](#); [Moriya \(1985\)](#). The analogy with QED is useful. There the local coupling $J_\mu A^\mu$ and the exchange of a photon between two charges gives the interaction $V_{\text{int}} = e^2/r$. In an itinerant magnet the local Hubbard interaction U couples the spin at site i to a particle–hole pair, which propagates to site j and couples again with strength U . The effective exchange coupling is therefore

$$J_{ij} \sim U^2 \chi_{ij}^0, \quad (1.1)$$

where χ_{ij}^0 is the bare particle–hole propagator (bubble) between sites i and j .



The particle–hole pair is created by the action of the spin-lowering operator $S_i^- = a_{i\downarrow}^\dagger a_{i\uparrow}$ on the ground state: an $\uparrow$ electron is removed from below the Fermi energy ϵ_F (leaving an $\uparrow$ hole) and a $\downarrow$ electron is placed above ϵ_F . The spin stiffness, discussed in Sec. 5, is the energy cost of twisting the spin configuration (“bending the rod”).

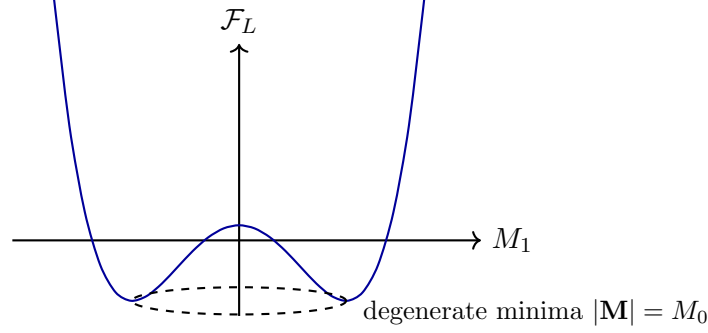
Topics. Many-body problem; Slater determinants; occupation-number representation and second quantization; creation–annihilation operators and their algebra; representation of the Coulomb interaction in the site basis (Sec. 6); coupled quantum wells and the extension to a lattice; the hopping term as the matrix element of the kinetic energy $-\hbar^2 \nabla^2 / 2m$ between neighbouring orbitals; the hopping Hamiltonian in the site basis $\{|i\rangle\}$ and its diagonalization using plane-wave states $|\mathbf{k}\rangle$ (Sec. 3).

2 Discrete vs. continuous symmetry, spontaneous symmetry breaking, Goldstone modes

We first discuss these ideas in a classical field-theory setting: the Landau theory of continuous phase transitions. For a two-component order parameter $\mathbf{M} = (M_1, M_2)$,

$$\mathcal{F}_L = \int d^d r \left[A(T - T_c) \mathbf{M}^2 + B (\mathbf{M}^2)^2 + C (\nabla M_1 \cdot \nabla M_1 + \nabla M_2 \cdot \nabla M_2) \right], \quad A, B, C > 0. \quad (2.1)$$

$\mathcal{F}_L$ is invariant under the continuous rotation $\mathbf{M} \rightarrow R(\theta)\mathbf{M}$ ($O(2)$ symmetry).



For $T < T_c$ the uniform minimum is at $|\mathbf{M}| = M_0$ with

$$M_0^2 = \frac{A(T_c - T)}{2B}. \quad (2.2)$$

All directions of $\mathbf{M}$ on the circle $|\mathbf{M}| = M_0$ are degenerate; the system chooses one of them (spontaneous symmetry breaking). Take $\mathbf{M}_0 = M_0 \hat{\mathbf{1}}$ and write

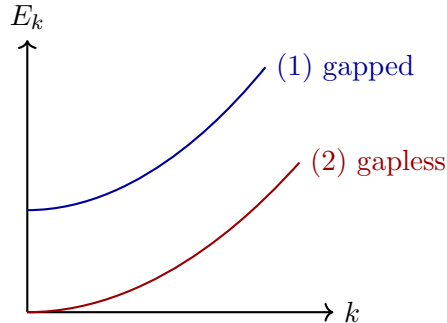
$$\mathbf{M}(\mathbf{r}) = (M_0 + m_1(\mathbf{r}), m_2(\mathbf{r})), \quad (2.3)$$

with m_1 the longitudinal and m_2 the transverse fluctuation. Writing $a \equiv A(T - T_c) < 0$ and expanding Eq. (2.1) to second order in $m_{1,2}$:

$$a\mathbf{M}^2 + B\mathbf{M}^4 = \text{const} + (a + 6BM_0^2) m_1^2 + (a + 2BM_0^2) m_2^2 + \dots = \text{const} + 2|a| m_1^2 + 0 \cdot m_2^2 + \dots \quad (2.4)$$

(the linear terms vanish at the minimum). Hence, in Fourier space,

$$\delta\mathcal{F}_L = \sum_{\mathbf{k}} \left[(2|a| + Ck^2) |m_1(\mathbf{k})|^2 + Ck^2 |m_2(\mathbf{k})|^2 \right]. \quad (2.5)$$



- (1) Longitudinal fluctuations (change of $|\mathbf{M}|$) cost a finite energy even at $k = 0$: gapped mode.
- (2) Transverse fluctuations (rotation of $\mathbf{M}$ along the degenerate circle) cost no energy at $k = 0$; the energy is $\propto k^2$: gapless mode.

Goldstone theorem. For a system possessing a continuous symmetry, there exist gapless (Goldstone) modes in the spontaneously broken-symmetry state (Goldstone et al., 1962). For a discrete symmetry (e.g. Ising, $M \rightarrow -M$) there is no such mode.

Domain wall (functional minimization). For a discrete (Ising-like) order parameter $M(x)$ in one dimension, minimization of $\mathcal{F}_L = \int dx [aM^2 + BM^4 + C(M')^2]$ gives the Euler–Lagrange equation (the same variational principle as in the brachistochrone problem)

$$C M'' = a M + 2B M^3. \quad (2.6)$$

With the boundary conditions $M(\pm\infty) = \pm M_0$ the solution is the domain wall

$$M(x) = M_0 \tanh(x/\xi), \quad \xi^2 = \frac{2C}{|a|}, \quad M_0^2 = \frac{|a|}{2B}, \quad (2.7)$$

as can be checked by using $(\tanh u)'' = -2 \tanh u (1 - \tanh^2 u)$.

3 Hopping term: tight-binding model

In the site (Wannier/atomic orbital) basis $\{|i\rangle\}$ (for second quantization see Fetter and Walecka 1971; Negele and Orland 1988) the nearest-neighbour hopping Hamiltonian is

$$\mathcal{H}_0 = -t \sum_{\langle ij \rangle} |i\rangle \langle j| + \text{H.c.} = -t \sum_i \sum_{\boldsymbol{\delta}} |i + \boldsymbol{\delta}\rangle \langle i|, \quad (3.1)$$

where $\langle ij \rangle$ runs over all nearest-neighbour pairs and $\boldsymbol{\delta}$ over the nearest-neighbour vectors of site i (square lattice, lattice spacing $a \equiv 1$).

Use the inverse Fourier transformation

$$|i\rangle = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{r}_i} |\mathbf{k}\rangle. \quad (3.2)$$

Then

$$\begin{aligned} \mathcal{H}_0 &= -t \sum_i \sum_{\boldsymbol{\delta}} \frac{1}{N} \sum_{\mathbf{k}, \mathbf{k}'} e^{i\mathbf{k} \cdot (\mathbf{r}_i + \boldsymbol{\delta})} e^{-i\mathbf{k}' \cdot \mathbf{r}_i} |\mathbf{k}\rangle \langle \mathbf{k}'| \\ &= -t \sum_{\boldsymbol{\delta}} \sum_{\mathbf{k}, \mathbf{k}'} \underbrace{\left(\frac{1}{N} \sum_i e^{i(\mathbf{k} - \mathbf{k}') \cdot \mathbf{r}_i} \right)}_{\delta_{\mathbf{k}\mathbf{k}'}} e^{i\mathbf{k} \cdot \boldsymbol{\delta}} |\mathbf{k}\rangle \langle \mathbf{k}'| \\ &= -t \sum_{\mathbf{k}} (e^{ik_x} + e^{-ik_x} + e^{ik_y} + e^{-ik_y}) |\mathbf{k}\rangle \langle \mathbf{k}| = \sum_{\mathbf{k}} \epsilon_{\mathbf{k}} |\mathbf{k}\rangle \langle \mathbf{k}|, \end{aligned} \quad (3.3)$$

so $\mathcal{H}_0$ is diagonalized, with band energies

$$\boxed{\epsilon_{\mathbf{k}} = -2t (\cos k_x + \cos k_y)} \quad (\text{square lattice}). \quad (3.4)$$

For the simple cubic lattice $\epsilon_{\mathbf{k}} = -2t(\cos k_x + \cos k_y + \cos k_z)$.

4 Hartree–Fock approximation for general magnetic ordering

The Hubbard model (Hubbard, 1963; Kanamori, 1963) is

$$\mathcal{H} = -t \sum_{\langle ij \rangle \sigma} (a_{i\sigma}^\dagger a_{j\sigma} + \text{H.c.}) + U \sum_i n_{i\uparrow} n_{i\downarrow}. \quad (4.1)$$

Define the spinor $\psi_i = (a_{i\uparrow}, a_{i\downarrow})^T$ and the spin operator

$$\mathbf{S}_i = \psi_i^\dagger \frac{\boldsymbol{\sigma}}{2} \psi_i, \quad S_i^\pm = S_i^x \pm i S_i^y, \quad S_i^+ = a_{i\uparrow}^\dagger a_{i\downarrow}, \quad S_i^- = a_{i\downarrow}^\dagger a_{i\uparrow}. \quad (4.2)$$

Decoupling

Write $n_{i\uparrow} n_{i\downarrow} = a_{i\uparrow}^\dagger a_{i\uparrow} a_{i\downarrow}^\dagger a_{i\downarrow}$ and keep all Hartree and Fock (exchange) contractions:

$$\begin{aligned} n_{i\uparrow} n_{i\downarrow} &\xrightarrow{\text{HF}} a_{i\uparrow}^\dagger a_{i\uparrow} \langle a_{i\downarrow}^\dagger a_{i\downarrow} \rangle + a_{i\downarrow}^\dagger a_{i\downarrow} \langle a_{i\uparrow}^\dagger a_{i\uparrow} \rangle - \langle n_{i\uparrow} \rangle \langle n_{i\downarrow} \rangle \\ &\quad - a_{i\uparrow}^\dagger a_{i\downarrow} \langle a_{i\downarrow}^\dagger a_{i\uparrow} \rangle - a_{i\downarrow}^\dagger a_{i\uparrow} \langle a_{i\uparrow}^\dagger a_{i\downarrow} \rangle + \langle S_i^+ \rangle \langle S_i^- \rangle. \end{aligned} \quad (4.3)$$

With total density n and magnetization m_i^z ,

$$\langle n_{i\uparrow} \rangle + \langle n_{i\downarrow} \rangle = n, \quad \langle n_{i\uparrow} \rangle - \langle n_{i\downarrow} \rangle = m_i^z \Rightarrow \langle n_{i\uparrow} \rangle = \frac{1}{2}(n + m_i^z), \quad \langle n_{i\downarrow} \rangle = \frac{1}{2}(n - m_i^z), \quad (4.4)$$

and $\langle S_i^z \rangle = m_i^z/2$. Substituting in (4.3):

$$\begin{aligned} n_{i\uparrow} n_{i\downarrow} &\xrightarrow{\text{HF}} -a_{i\uparrow}^\dagger a_{i\uparrow} \langle S_i^z \rangle + a_{i\downarrow}^\dagger a_{i\downarrow} \langle S_i^z \rangle - a_{i\uparrow}^\dagger a_{i\downarrow} \langle S_i^- \rangle - a_{i\downarrow}^\dagger a_{i\uparrow} \langle S_i^+ \rangle \\ &\quad + \frac{n}{2} (a_{i\uparrow}^\dagger a_{i\uparrow} + a_{i\downarrow}^\dagger a_{i\downarrow}) - \langle n_{i\uparrow} \rangle \langle n_{i\downarrow} \rangle + \langle S_i^+ \rangle \langle S_i^- \rangle \\ &= - \begin{pmatrix} a_{i\uparrow}^\dagger & a_{i\downarrow}^\dagger \end{pmatrix} \begin{pmatrix} \langle S_i^z \rangle & \langle S_i^- \rangle \\ \langle S_i^+ \rangle & -\langle S_i^z \rangle \end{pmatrix} \begin{pmatrix} a_{i\uparrow} \\ a_{i\downarrow} \end{pmatrix} + \frac{n}{2} \psi_i^\dagger \psi_i - \frac{n^2}{4} + \langle S_i^x \rangle^2 + \langle S_i^y \rangle^2 + \langle S_i^z \rangle^2. \end{aligned} \quad (4.5)$$

Two checks were used here:

$$\boldsymbol{\sigma} \cdot \langle \mathbf{S}_i \rangle = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \langle S_i^x \rangle + \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \langle S_i^y \rangle + \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \langle S_i^z \rangle = \begin{pmatrix} \langle S_i^z \rangle & \langle S_i^- \rangle \\ \langle S_i^+ \rangle & -\langle S_i^z \rangle \end{pmatrix}, \quad (4.6)$$

$$\langle S_i^+ \rangle \langle S_i^- \rangle = (\langle S_i^x \rangle + i \langle S_i^y \rangle) (\langle S_i^x \rangle - i \langle S_i^y \rangle) = \langle S_i^x \rangle^2 + \langle S_i^y \rangle^2, \quad (4.7)$$

and $-\langle n_{i\uparrow} \rangle \langle n_{i\downarrow} \rangle = -n^2/4 + \langle S_i^z \rangle^2$. Therefore

$$n_{i\uparrow} n_{i\downarrow} \xrightarrow{\text{HF}} -\psi_i^\dagger [\boldsymbol{\sigma} \cdot \langle \mathbf{S}_i \rangle] \psi_i + \dots = -2 \mathbf{S}_i \cdot \langle \mathbf{S}_i \rangle + \dots, \quad (4.8)$$

where “...” denotes the density term and c -number constants.

Result

The Hubbard interaction term in the HF approximation:

$$U n_{i\uparrow} n_{i\downarrow} \xrightarrow{\text{HF}} -2 \mathbf{S}_i \cdot (U \langle \mathbf{S}_i \rangle) + \dots = -\boldsymbol{\sigma}_i \cdot \boldsymbol{\Delta}_i + \dots = -\psi_i^\dagger [\boldsymbol{\sigma} \cdot \boldsymbol{\Delta}_i] \psi_i + \dots \quad (4.9)$$

with the exchange field

$$\boldsymbol{\Delta}_i = U \langle \mathbf{S}_i \rangle, \quad 2\boldsymbol{\Delta}_i = \mathbf{m}_i U, \quad (4.10)$$

where $\boldsymbol{\sigma}_i \equiv \psi_i^\dagger \boldsymbol{\sigma} \psi_i$ is the (Pauli-matrix) spin-density operator. The orientation of $\boldsymbol{\Delta}_i$ is arbitrary: this describes general (ferro, antiferro, spiral, ...) magnetic ordering.

The same result is obtained starting from the purely local interaction $-U \mathbf{S}_i \cdot \mathbf{S}_i$ in the mean-field approximation:

$$-U \mathbf{S}_i \cdot \mathbf{S}_i \xrightarrow{\text{MF}} -2 \mathbf{S}_i \cdot (U \langle \mathbf{S}_i \rangle) + U \langle \mathbf{S}_i \rangle \cdot \langle \mathbf{S}_i \rangle. \quad (4.11)$$

5 Spin fluctuations in correlated electron systems

Outline of the approach:

- I. Spontaneous symmetry breaking: HF approximation, $\Delta_i = U\langle \mathbf{S}_i \rangle$ (Sec. 4).
- II. Transverse spin fluctuations (spin waves): ladder-sum approximation.
- III. Self-energy corrections: electron coupling with spin fluctuations.
- IV. Quantum corrections to spin waves: self-energy and vertex corrections.

This lecture treats item II.

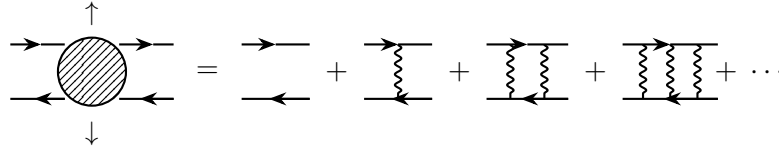
Transverse spin susceptibility

Assume, without loss of generality, ordering along $\hat{z}$ ($|\uparrow\uparrow\uparrow\dots\rangle$). In analogy with the classical field theory, consider (following [Izuyama et al. 1963](#); [Moriya 1985](#)) the transverse (x, y) fluctuations through the time-ordered propagator

$$\chi_{ij}^{-+}(t, t') = \langle \Psi_{\text{gnd}} | T [S_i^-(t) S_j^+(t')] | \Psi_{\text{gnd}} \rangle, \quad S_i^- = S_i^x - iS_i^y, \quad S_j^+ = S_j^x + iS_j^y, \quad (5.1)$$

where T is the time-ordering operator.

Ladder-sum approximation. The particle-hole ($\uparrow$ particle, $\downarrow$ hole) propagator is summed with repeated interactions U between the two lines:



$$\text{i.e. } \chi = \chi^0 + \chi^0 U \chi^0 + \chi^0 U \chi^0 U \chi^0 + \dots$$

Bare particle-hole propagator for the saturated ferromagnet

In the HF state the bands are spin split,

$$\epsilon_{\mathbf{k}}^{\uparrow} = \epsilon_{\mathbf{k}} - \Delta, \quad \epsilon_{\mathbf{k}}^{\downarrow} = \epsilon_{\mathbf{k}} + \Delta, \quad 2\Delta = mU. \quad (5.2)$$

Consider the *saturated ferromagnet*: the $\downarrow$ band is empty and the $\uparrow$ band contains all electrons, $n_{\uparrow} = m$ (states below ϵ_F marked $\ominus$, empty states marked $\oplus$). The bare particle-hole propagator, with a $\uparrow$ line ($\mathbf{k}', \omega'$) and a $\downarrow$ line ($\mathbf{k}' - \mathbf{q}, \omega' - \omega$), is

$$\begin{aligned} \chi^0(\mathbf{q}, \omega) &= \sum_{\mathbf{k}'} i \int_{-\infty}^{\infty} \frac{d\omega'}{2\pi} G_0^{\uparrow}(\mathbf{k}', \omega') G_0^{\downarrow}(\mathbf{k}' - \mathbf{q}, \omega' - \omega) \\ &= \sum_{\mathbf{k}'}^{\ominus} i \int \frac{d\omega'}{2\pi} \left(\frac{1}{\omega' - \epsilon_{\mathbf{k}'}^{\uparrow} - i\eta} \right) \left(\frac{1}{\omega' - \omega - \epsilon_{\mathbf{k}' - \mathbf{q}}^{\downarrow} + i\eta} \right). \end{aligned} \quad (5.3)$$

$G_0^\uparrow$ has both advanced (occupied, $\ominus$) and retarded (empty, $\oplus$) components. For an empty $\uparrow$ state both poles lie in the lower half-plane and the ω' integral (closing the contour in the upper half-plane) vanishes. Hence only the advanced part contributes. Closing in the upper half-plane around $\omega' = \epsilon_{\mathbf{k}'}^\uparrow + i\eta$:

$$\begin{aligned}\chi^0(\mathbf{q}, \omega) &= \sum_{\mathbf{k}'}^{\ominus} \frac{i}{2\pi} 2\pi i \frac{1}{\epsilon_{\mathbf{k}'}^\uparrow - \epsilon_{\mathbf{k}'-\mathbf{q}}^\downarrow - \omega + i\eta} = \sum_{\mathbf{k}'}^{\ominus} \frac{1}{\epsilon_{\mathbf{k}'-\mathbf{q}}^\downarrow - \epsilon_{\mathbf{k}'}^\uparrow + \omega - i\eta} \\ &= \sum_{\mathbf{k}'}^{\ominus} \frac{1}{2\Delta + \epsilon_{\mathbf{k}'-\mathbf{q}} - \epsilon_{\mathbf{k}'} + \omega - i\eta} = \sum_{\mathbf{k}'}^{\ominus} \frac{1}{2\Delta \left[1 + \frac{\epsilon_{\mathbf{k}'-\mathbf{q}} - \epsilon_{\mathbf{k}'}}{2\Delta} + \frac{\omega}{2\Delta} - \frac{i\eta}{2\Delta} \right]}.\end{aligned}\quad (5.4)$$

To leading order in $t/2\Delta$ (strong coupling),

$$\begin{aligned}\chi^0(\mathbf{q}, \omega) &\simeq \sum_{\mathbf{k}'}^{\ominus} \frac{1}{2\Delta} \left[1 - \frac{\epsilon_{\mathbf{k}'-\mathbf{q}} - \epsilon_{\mathbf{k}'}}{2\Delta} - \frac{\omega}{2\Delta} + \frac{i\eta}{2\Delta} \right] \\ &= \frac{m}{2\Delta} \left[1 - \frac{\omega}{2\Delta} + \frac{i\eta}{2\Delta} \right] - \left(\frac{1}{2\Delta} \right)^2 \sum_{\mathbf{k}'}^{\ominus} (\epsilon_{\mathbf{k}'-\mathbf{q}} - \epsilon_{\mathbf{k}'}),\end{aligned}\quad (5.5)$$

using $\sum_{\mathbf{k}'}^{\ominus} 1 = n_\uparrow = m$ (sums normalized per site), and $m/2\Delta = 1/U$.

Simple cubic lattice. With $\epsilon_{\mathbf{k}'} = -2t(\cos k'_x + \cos k'_y + \cos k'_z)$,

$$\epsilon_{\mathbf{k}'-\mathbf{q}} - \epsilon_{\mathbf{k}'} = -2t \sum_{\mu=x,y,z} \left[\cos k'_\mu (\cos q_\mu - 1) + \sin k'_\mu \sin q_\mu \right]. \quad (5.6)$$

By cubic symmetry, with averages $\langle \dots \rangle$ normalized over the number of occupied states,

$$\langle \cos k'_x \rangle = \langle \cos k'_y \rangle = \langle \cos k'_z \rangle, \quad \langle \sin k'_x \rangle = \langle \sin k'_y \rangle = \langle \sin k'_z \rangle = 0, \quad (5.7)$$

so that

$$\sum_{\mathbf{k}'}^{\ominus} (\epsilon_{\mathbf{k}'-\mathbf{q}} - \epsilon_{\mathbf{k}'}) = +2t \langle \cos k'_x \rangle 3(1 - \gamma_{\mathbf{q}}) m, \quad \gamma_{\mathbf{q}} \equiv \frac{1}{3}(\cos q_x + \cos q_y + \cos q_z). \quad (5.8)$$

Hence

$$\chi^0(\mathbf{q}, \omega) = \frac{1}{U} \left[1 - \frac{\omega}{2\Delta} + \frac{i\eta}{2\Delta} - \frac{U}{(2\Delta)^2} m \omega_{\mathbf{q}} \right], \quad \omega_{\mathbf{q}} \equiv (zt) \langle \cos k'_x \rangle (1 - \gamma_{\mathbf{q}}), \quad (5.9)$$

where z is the lattice coordination number ($z = 6$ here).

Ladder sum: spin-wave propagator

$$\chi^{-+}(\mathbf{q}, \omega) = \frac{\chi^0(\mathbf{q}, \omega)}{1 - U\chi^0(\mathbf{q}, \omega)} = \frac{1}{[\chi^0(\mathbf{q}, \omega)]^{-1} - U} \simeq \frac{U^{-1}}{\frac{\omega}{2\Delta} - \frac{i\eta}{2\Delta} + \frac{Um}{(2\Delta)^2} \omega_{\mathbf{q}}}. \quad (5.10)$$

Using $2\Delta = mU$:

$$\chi^{-+}(\mathbf{q}, \omega) = \frac{2\Delta/U = m}{\omega + \omega_{\mathbf{q}} - i\eta} \quad (\text{purely advanced propagator}), \quad (5.11)$$

$$\chi^{+-}(\mathbf{q}, \omega) = \chi^{-+}(-\mathbf{q}, -\omega) = \frac{-m}{\omega - \omega_{\mathbf{q}} + i\eta} \quad (\text{purely retarded propagator}). \quad (5.12)$$

The pole at $\omega = \omega_{\mathbf{q}}$ is the spin-wave (magnon) energy. It vanishes as $\mathbf{q} \rightarrow 0$: this is the Goldstone mode of the broken spin-rotation symmetry.

Small- q expansion: spin stiffness

We now evaluate $\chi^0(\mathbf{q}, \omega)$ to quadratic order in q but keeping all orders in $(\epsilon_{\mathbf{k}-\mathbf{q}} - \epsilon_{\mathbf{k}})/2\Delta$ that contribute at $O(q^2)$ (here $\mathbf{k}' \rightarrow \mathbf{k}$). Expand

$$\epsilon_{\mathbf{k}-\mathbf{q}} \simeq \epsilon_{\mathbf{k}} - \mathbf{q} \cdot \nabla_{\mathbf{k}} \epsilon_{\mathbf{k}} + \frac{1}{2}(\mathbf{q} \cdot \nabla_{\mathbf{k}})^2 \epsilon_{\mathbf{k}} + \dots \quad (5.13)$$

and substitute in (5.4). With $x \equiv (\epsilon_{\mathbf{k}-\mathbf{q}} - \epsilon_{\mathbf{k}})/2\Delta$ and $1/(1+x+w) \simeq 1-w-x+x^2$, the term linear in $\nabla_{\mathbf{k}} \epsilon_{\mathbf{k}}$ averages to zero, and up to $O(q^2)$:

$$\begin{aligned} \chi^0(\mathbf{q}, \omega) &= \frac{m}{2\Delta} \left[\left(1 - \frac{\omega}{2\Delta} + \frac{i\eta}{2\Delta}\right) - \frac{1}{2} \frac{1}{2\Delta} \langle (\mathbf{q} \cdot \nabla_{\mathbf{k}})^2 \epsilon_{\mathbf{k}} \rangle + \left(\frac{1}{2\Delta}\right)^2 \langle (\mathbf{q} \cdot \nabla_{\mathbf{k}} \epsilon_{\mathbf{k}})^2 \rangle \right] \\ &= \frac{1}{U} \left[\left(1 - \frac{\omega}{2\Delta} + \frac{i\eta}{2\Delta}\right) - \frac{1}{2} \frac{1}{2\Delta} q^2 \langle \frac{\partial^2 \epsilon_{\mathbf{k}}}{\partial k_x^2} \rangle + \left(\frac{1}{2\Delta}\right)^2 q^2 \langle \left(\frac{\partial \epsilon_{\mathbf{k}}}{\partial k_x}\right)^2 \rangle \right], \end{aligned} \quad (5.14)$$

with $q^2 = q_x^2 + q_y^2 + q_z^2$ and $\langle \dots \rangle$ the normalized sum over occupied $\mathbf{k}$ states. Here cubic symmetry was used again:

$$\langle \frac{\partial^2 \epsilon_{\mathbf{k}}}{\partial k_x^2} \rangle = \langle \frac{\partial^2 \epsilon_{\mathbf{k}}}{\partial k_y^2} \rangle = \langle \frac{\partial^2 \epsilon_{\mathbf{k}}}{\partial k_z^2} \rangle, \quad \langle \frac{\partial^2 \epsilon_{\mathbf{k}}}{\partial k_x \partial k_y} \rangle = 0, \quad \langle \frac{\partial \epsilon_{\mathbf{k}}}{\partial k_x} \frac{\partial \epsilon_{\mathbf{k}}}{\partial k_y} \rangle = 0, \quad (5.15)$$

and similarly for $\langle (\partial \epsilon_{\mathbf{k}} / \partial k_{\mu})^2 \rangle$.

Hence, in the ladder-sum approximation,

$$\chi^{-+}(\mathbf{q}, \omega) = \frac{\chi^0(\mathbf{q}, \omega)}{1 - U\chi^0(\mathbf{q}, \omega)} = \frac{2\Delta/U = m}{\omega + Dq^2 - i\eta}, \quad (5.16)$$

with the spin stiffness constant

$$D = \frac{1}{2} \langle \frac{\partial^2 \epsilon_{\mathbf{k}}}{\partial k_x^2} \rangle - \frac{1}{2\Delta} \langle \left(\frac{\partial \epsilon_{\mathbf{k}}}{\partial k_x}\right)^2 \rangle. \quad (5.17)$$

Physical interpretation.

- The first term is of $O(t)$: the increase of the delocalization (kinetic) energy on creating a twisted spin configuration.
- The second term is of $O(t^2/U)$: the decrease of the exchange energy for the twisted spin configuration.

The net spin stiffness involves a competition between the two effects, whose magnitudes depend on the band structure ($\epsilon_{\mathbf{k}}$) and the band filling. The second term vanishes in the strong-coupling limit $t/2\Delta \rightarrow 0$.

Remark. For the simple cubic band, $\frac{1}{2}\langle\partial^2\epsilon_{\mathbf{k}}/\partial k_x^2\rangle = t\langle\cos k_x\rangle$, and for small q ,

$$\omega_{\mathbf{q}} = 6t\langle\cos k_x\rangle \frac{1}{3} \sum_{\mu} (1 - \cos q_{\mu}) \simeq t\langle\cos k_x\rangle q^2.$$

Thus the leading-order (in $t/2\Delta$) result of the previous subsection agrees with the first term of Eq. (5.17).

Extension to a multi-orbital system

E.g. a 3d system with yz , xz , xy , $3z^2 - r^2$, $x^2 - y^2$ orbitals.

$$\mathcal{H} = \sum_{\langle ij \rangle} \sum_{\mu, \nu} \sum_{\sigma} \psi_{i\mu\sigma}^{\dagger} t_{ij}^{\mu\nu} \psi_{j\nu\sigma} + \sum_{i\mu} U_{\mu} n_{i\mu\uparrow} n_{i\mu\downarrow} - 2J \sum_{i; \mu < \nu} \mathbf{S}_{i\mu} \cdot \mathbf{S}_{i\nu}, \quad (5.18)$$

where the last term is the Hund's rule coupling (see the Coulomb interaction matrix elements in Sec. 6; each orbital pair is counted once).

HF approximation.

$$\boxed{2\Delta_{\mu} = U_{\mu} m_{\mu} + J \sum_{\nu \neq \mu} m_{\nu}}, \quad (5.19)$$

$$\mathcal{H}_{\text{HF}}(\mathbf{k}) = \sum_{\mathbf{k}, \sigma, \mu, \nu} \psi_{\mathbf{k}\sigma\mu}^{\dagger} \left[\epsilon_{\mathbf{k}}^{\mu\nu} - \sigma \Delta_{\mu} \delta_{\mu\nu} \right] \psi_{\mathbf{k}\sigma\nu}. \quad (5.20)$$

Ladder sum. The bare propagator $[\chi^0(\mathbf{q}, \omega)]^{\mu\nu}$ now carries orbital indices ($\uparrow$ line $\mathbf{k}'$, ω' from μ to ν ; $\downarrow$ line $\mathbf{k} - \mathbf{q}$, $\omega' - \omega$ from ν to μ). In matrix form,

$$[\chi^{-+}(\mathbf{q}, \omega)] = \frac{[\chi^0(\mathbf{q}, \omega)]}{\mathbb{I} - [U][\chi^0(\mathbf{q}, \omega)]}, \quad U_{\mu\nu} = \begin{cases} U_{\mu}, & \mu = \nu, \\ J, & \mu \neq \nu. \end{cases} \quad (5.21)$$

6 Lattice electrons: Coulomb interaction matrix elements

The on-site and inter-site parameters introduced here are those of Hubbard (1963); Kanamori (1963).

$$V_{\text{int}} = \frac{1}{2} \sum_{\sigma, \sigma'} \iint d\mathbf{r} d\mathbf{r}' \psi_{\sigma}^{\dagger}(\mathbf{r}) \psi_{\sigma'}^{\dagger}(\mathbf{r}') \frac{e^2}{|\mathbf{r} - \mathbf{r}'|} \psi_{\sigma'}(\mathbf{r}') \psi_{\sigma}(\mathbf{r}). \quad (6.1)$$

Substitute for the field operator

$$\psi_{\sigma}(\mathbf{r}) = \sum_i \phi_i(\mathbf{r}) a_{i\sigma}, \quad (6.2)$$

where $\phi_i(\mathbf{r})$ is the atomic-orbital wavefunction at lattice site i . Then

$$V_{\text{int}} = \frac{1}{2} \sum_{\sigma, \sigma'} \sum_{i, j, k, l} V_{ijkl} a_{i\sigma}^{\dagger} a_{j\sigma'}^{\dagger} a_{k\sigma'} a_{l\sigma}, \quad V_{ijkl} = \iint d\mathbf{r} d\mathbf{r}' \phi_i^*(\mathbf{r}) \phi_j^*(\mathbf{r}') \frac{e^2}{|\mathbf{r} - \mathbf{r}'|} \phi_k(\mathbf{r}') \phi_l(\mathbf{r}). \quad (6.3)$$

We separate out the different matrix elements.

(1) $i = j = k = l$: intra-orbital Coulomb interaction. The two spins must be opposite ($\sigma' = \bar{\sigma}$):

$$V_{\text{int}} = U \sum_i a_{i\uparrow}^\dagger a_{i\downarrow}^\dagger a_{i\downarrow} a_{i\uparrow} = U \sum_i n_{i\uparrow} n_{i\downarrow}, \quad U = \iint \phi_i^*(\mathbf{r}) \phi_i^*(\mathbf{r}') \frac{e^2}{|\mathbf{r} - \mathbf{r}'|} \phi_i(\mathbf{r}') \phi_i(\mathbf{r}). \quad (6.4)$$

(2) $k = j, l = i$: direct term (inter-site Coulomb interaction).

$$V_{ijji} \equiv V = \iint \phi_i^*(\mathbf{r}) \phi_j^*(\mathbf{r}') \frac{e^2}{|\mathbf{r} - \mathbf{r}'|} \phi_j(\mathbf{r}') \phi_i(\mathbf{r}). \quad (6.5)$$

The spins can be same or different. Using $\sum_{i,j} = 2 \sum_{\langle ij \rangle}$,

$$V_{\text{int}} = 2 \times \frac{V}{2} \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} a_{i\sigma}^\dagger a_{j\sigma'}^\dagger a_{j\sigma'} a_{i\sigma} = V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} n_{i\sigma} n_{j\sigma'} = 4V \sum_{\langle ij \rangle} n_i n_j, \quad (6.6)$$

where in this section $n_i \equiv \frac{1}{2}(n_{i\uparrow} + n_{i\downarrow})$.

(3) $k = i, l = j$: exchange term.

$$V_{ijij} \equiv J = \iint \phi_i^*(\mathbf{r}) \phi_j^*(\mathbf{r}') \frac{e^2}{|\mathbf{r} - \mathbf{r}'|} \phi_i(\mathbf{r}') \phi_j(\mathbf{r}). \quad (6.7)$$

If $\sigma' = \sigma$,

$$V_{\text{int}} = J \sum_{\langle ij \rangle} \sum_{\sigma} a_{i\sigma}^\dagger a_{j\sigma}^\dagger a_{i\sigma} a_{j\sigma} = -J \sum_{\langle ij \rangle} \sum_{\sigma} (a_{i\sigma}^\dagger a_{i\sigma})(a_{j\sigma}^\dagger a_{j\sigma}). \quad (6.8)$$

Terms (2) and (3) together: $\sum_{\langle ij \rangle} \sum_{\sigma\sigma'} [V - J\delta_{\sigma\sigma'}] n_{i\sigma} n_{j\sigma'}$.

(4) $\sigma' \neq \sigma$: transverse part of the exchange term.

$$\begin{aligned} V_{\text{int}} &= J \sum_{\langle ij \rangle} \sum_{\sigma} a_{i\sigma}^\dagger a_{j\bar{\sigma}}^\dagger a_{i\bar{\sigma}} a_{j\sigma} = J \sum_{\langle ij \rangle} (a_{i\uparrow}^\dagger a_{j\downarrow}^\dagger a_{i\downarrow} a_{j\uparrow} + a_{i\downarrow}^\dagger a_{j\uparrow}^\dagger a_{i\uparrow} a_{j\downarrow}) \\ &= -J \sum_{\langle ij \rangle} [(a_{i\uparrow}^\dagger a_{i\downarrow})(a_{j\downarrow}^\dagger a_{j\uparrow}) + (a_{i\downarrow}^\dagger a_{i\uparrow})(a_{j\uparrow}^\dagger a_{j\downarrow})] = -J \sum_{\langle ij \rangle} (S_i^+ S_j^- + S_i^- S_j^+) \\ &= -2J \sum_{\langle ij \rangle} [S_i^x S_j^x + S_i^y S_j^y]. \end{aligned} \quad (6.9)$$

(5) Rewrite (3).

$$\begin{aligned} -J[n_{i\uparrow} n_{j\uparrow} + n_{i\downarrow} n_{j\downarrow}] &= -2J \left[\left(\frac{n_{i\uparrow} - n_{i\downarrow}}{2} \right) \left(\frac{n_{j\uparrow} - n_{j\downarrow}}{2} \right) + \left(\frac{n_{i\uparrow} + n_{i\downarrow}}{2} \right) \left(\frac{n_{j\uparrow} + n_{j\downarrow}}{2} \right) \right] \\ &= -2J [S_i^z S_j^z + n_i n_j]. \end{aligned} \quad (6.10)$$

Together, (4) and (5) (the exchange terms) give

$$-2J \sum_{\langle ij \rangle} [\mathbf{S}_i \cdot \mathbf{S}_j + n_i n_j]. \quad (6.11)$$

The same calculation with two orbitals α, β on the same site ($i, j \rightarrow \alpha, \beta$) gives the Hund's coupling term $-2J \sum_{\alpha < \beta} \mathbf{S}_\alpha \cdot \mathbf{S}_\beta$.

(6) Pair-hopping term.

$$V_{iijj} \equiv J' = \iint \phi_i^*(\mathbf{r}) \phi_i^*(\mathbf{r}') \frac{e^2}{|\mathbf{r} - \mathbf{r}'|} \phi_j(\mathbf{r}') \phi_j(\mathbf{r}), \quad (6.12)$$

with $\sigma' = \bar{\sigma}$ (a pair of electrons on the same site i or j):

$$V_{\text{int}} = J' \sum_{\langle ij \rangle} \sum_{\sigma} a_{i\sigma}^\dagger a_{i\bar{\sigma}}^\dagger a_{j\bar{\sigma}} a_{j\sigma}. \quad (6.13)$$

7 The strong-coupling limit: expansion in powers of t

Hubbard model, 1 electron per site $\xrightarrow{U/t \gg 1}$ quantum Heisenberg antiferromagnet ($S = \frac{1}{2}$):

$$\mathcal{H}_{\text{eff}} = J \sum_{\langle ij \rangle} \left(\mathbf{S}_i \cdot \mathbf{S}_j - \frac{1}{4} \right), \quad J \equiv \frac{4t^2}{U}.$$

The kinetic-exchange mechanism goes back to [Anderson \(1959\)](#); the systematic t/U expansion is given in [Chao et al. \(1977\)](#); [MacDonald et al. \(1988\)](#).

Unperturbed problem

Split the Hubbard Hamiltonian as

$$\mathcal{H}_0 = U \sum_i n_{i\uparrow} n_{i\downarrow} \quad (\text{unperturbed part}), \quad (7.1)$$

$$\mathcal{H}' = -t \sum_{\langle ij \rangle \sigma} (a_{i\sigma}^\dagger a_{j\sigma} + a_{j\sigma}^\dagger a_{i\sigma}) \quad (\text{nearest-neighbour hopping, treated as perturbation}). \quad (7.2)$$

For a single site i , the eigenstates of $\mathcal{H}_0(i)$ are $|0\rangle, |\uparrow\rangle, |\downarrow\rangle, |\uparrow\downarrow\rangle$ with energies $0, 0, 0, U$. All sites are decoupled at the $\mathcal{H}_0$ level.

The N -particle ground state has one electron at every site, in any spin state: it is 2^N -fold degenerate. We denote it by $|\Psi_g^{(0)}\rangle$, with $E_g^{(0)} = 0$.

Perturbation theory

First order.

$$\delta E_g^{(1)} = \langle \Psi_g^{(0)} | \mathcal{H}' | \Psi_g^{(0)} \rangle = 0, \quad (7.3)$$

since the hopping term moves an electron to a neighbouring site, so that in $\mathcal{H}' |\Psi_g^{(0)}\rangle$ one site is necessarily doubly occupied. There is no first-order correction.

Second order.

$$\delta E_g^{(2)} = \sum_{m \neq g} \frac{\langle \Psi_g^{(0)} | \mathcal{H}' | \Psi_m \rangle \langle \Psi_m | \mathcal{H}' | \Psi_g^{(0)} \rangle}{E_g^{(0)} - E_m}, \quad (7.4)$$

where the sum runs over all intermediate (excited) states. Since $E_m > E_g^{(0)}$, the second-order correction to the ground-state energy is always negative.

Since $\mathcal{H}' |\Psi_m\rangle$ must fall back into the ground-state manifold (the 2^N states), $|\Psi_m\rangle$ can have at most one doubly occupied and one empty site, and these must be nearest neighbours. Hence

$$E_m = U \quad (\text{intermediate-state energy}), \quad (7.5)$$

and the effective Hamiltonian acting within the ground-state manifold is

$$\mathcal{H}_{\text{eff}}^{(2)} = -\frac{1}{U} \sum_{m \neq g} \mathcal{H}' |\Psi_m\rangle \langle \Psi_m| \mathcal{H}'. \quad (7.6)$$

Evaluation for one bond

Take the intermediate state with site j doubly occupied and site i empty (other singly occupied sites not shown):

$$|\Psi_m\rangle = \left[\begin{array}{c} \uparrow\downarrow \\ j \end{array} \text{---} \begin{array}{c} \circ \\ i \end{array} \right] = a_{j\uparrow}^\dagger a_{j\downarrow}^\dagger |0\rangle, \quad (7.7)$$

where $|0\rangle \equiv |0;0_j\rangle$ has the sites i, j empty. Consider

$$\mathcal{H}' |\Psi_m\rangle = -t \sum_{\langle i'j' \rangle \sigma} \left(\underbrace{a_{i'\sigma}^\dagger a_{j'\sigma}}_{\text{I}} + \underbrace{a_{j'\sigma}^\dagger a_{i'\sigma}}_{\text{II}} \right) a_{j\uparrow}^\dagger a_{j\downarrow}^\dagger |0\rangle. \quad (7.8)$$

- Term I: must have $j' = j, i' = i$. For $\sigma = \uparrow$: $-t a_{i\uparrow}^\dagger a_{j\downarrow}^\dagger |0\rangle$. For $\sigma = \downarrow$: $-t (-a_{i\downarrow}^\dagger a_{j\uparrow}^\dagger) |0\rangle$.
- Term II: must have $i' = j, j' = i$; similarly (for $\sigma = \uparrow$ and $\downarrow$) it gives $-t(a_{i\uparrow}^\dagger a_{j\downarrow}^\dagger - a_{i\downarrow}^\dagger a_{j\uparrow}^\dagger) |0\rangle$.

Since the bond (i, j) is fixed, only one of I and II actually contributes. Hence

$$\mathcal{H}' |\Psi_m\rangle = -t(a_{i\uparrow}^\dagger a_{j\downarrow}^\dagger - a_{i\downarrow}^\dagger a_{j\uparrow}^\dagger) |0\rangle, \quad (7.9)$$

a spin singlet on the bond. Then

$$\begin{aligned} \mathcal{H}' |\Psi_m\rangle \langle \Psi_m| \mathcal{H}' &= t^2 (a_{i\uparrow}^\dagger a_{j\downarrow}^\dagger - a_{i\downarrow}^\dagger a_{j\uparrow}^\dagger) |0\rangle \langle 0| (a_{j\downarrow} a_{i\uparrow} - a_{j\uparrow} a_{i\downarrow}) \\ &= t^2 \left[n_{i\uparrow} n_{j\downarrow} + n_{i\downarrow} n_{j\uparrow} - (S_i^+ S_j^- + S_i^- S_j^+) \right], \end{aligned} \quad (7.10)$$

where in the second line we used that the operator acts only within the subspace in which sites i and j are each singly occupied; there, e.g., $a_{i\uparrow}^\dagger a_{j\downarrow}^\dagger |0\rangle \langle 0| a_{j\downarrow} a_{i\uparrow} = |\uparrow_i \downarrow_j\rangle \langle \uparrow_i \downarrow_j| = n_{i\uparrow} n_{j\downarrow}$.

Spin form

In the singly occupied subspace ($n_{i\uparrow} + n_{i\downarrow} = 1$),

$$S_i^z S_j^z = \frac{1}{4} (n_{i\uparrow} - n_{i\downarrow})(n_{j\uparrow} - n_{j\downarrow}) = \frac{1}{4} (n_{i\uparrow} n_{j\uparrow} + n_{i\downarrow} n_{j\downarrow} - n_{i\uparrow} n_{j\downarrow} - n_{i\downarrow} n_{j\uparrow}), \quad (7.11)$$

$$\frac{1}{4} = \frac{1}{4} (n_{i\uparrow} + n_{i\downarrow})(n_{j\uparrow} + n_{j\downarrow}) = \frac{1}{4} (n_{i\uparrow} n_{j\uparrow} + n_{i\downarrow} n_{j\downarrow} + n_{i\uparrow} n_{j\downarrow} + n_{i\downarrow} n_{j\uparrow}). \quad (7.12)$$

Therefore, by subtracting,

$$2 \left(S_i^z S_j^z - \frac{1}{4} \right) = -(n_{i\uparrow} n_{j\downarrow} + n_{i\downarrow} n_{j\uparrow}). \quad (7.13)$$

Furthermore,

$$S_i^+ S_j^- + S_i^- S_j^+ = (S_i^x + iS_i^y)(S_j^x - iS_j^y) + (S_i^x - iS_i^y)(S_j^x + iS_j^y) = 2(S_i^x S_j^x + S_i^y S_j^y). \quad (7.14)$$

We obtain

$$\mathcal{H}' |\Psi_m\rangle \langle \Psi_m| \mathcal{H}' = -2t^2 \left(\mathbf{S}_i \cdot \mathbf{S}_j - \frac{1}{4} \right). \quad (7.15)$$

For each pair of sites (i, j) there are two intermediate states (double occupation on j with i empty, and on i with j empty). Thus

$$\mathcal{H}_{\text{eff}}^{(2)} = \sum_m \frac{2t^2}{U} \left(\mathbf{S}_i \cdot \mathbf{S}_j - \frac{1}{4} \right) = \sum_{\langle ij \rangle} \frac{4t^2}{U} \left(\mathbf{S}_i \cdot \mathbf{S}_j - \frac{1}{4} \right), \quad J = \frac{4t^2}{U} \text{ (exchange energy)}, \quad (7.16)$$

where

$$\mathbf{S}_i = \begin{pmatrix} a_{i\uparrow}^\dagger & a_{i\downarrow}^\dagger \end{pmatrix} \frac{1}{2} \hbar \boldsymbol{\sigma} \begin{pmatrix} a_{i\uparrow} \\ a_{i\downarrow} \end{pmatrix} = \psi_i^\dagger \frac{1}{2} \hbar \boldsymbol{\sigma} \psi_i \quad (7.17)$$

are the $S = \frac{1}{2}$ spin angular momentum operators.

Remark. The exchange $J = 4t^2/U$ is antiferromagnetic ($J > 0$): only an antiparallel (singlet) pair can lower its energy by virtual hopping, since for parallel spins the hop is forbidden by the Pauli principle. This is the strong-coupling counterpart of $J_{ij} \sim U^2 \chi_{ij}^0$ in Eq. (1.1).

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